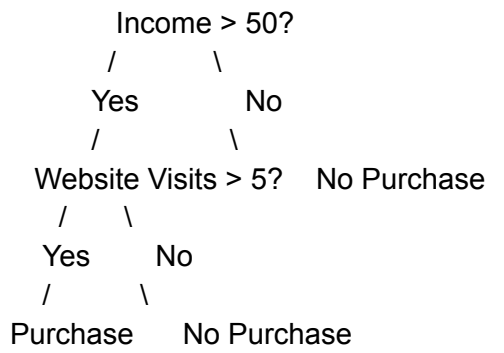
1. What is a Decision Tree?
A Decision Tree is a supervised machine learning algorithm that makes predictions by repeatedly dividing data based on conditions on input variables.
It looks similar to a tree:



The model starts at the top and moves through a series of questions until it reaches a final prediction.

Why is it called a Decision Tree?
It is called a Decision Tree because the model looks like a tree and makes a sequence of decisions.
For example:
Is income greater than ₹50 lakh?
If Yes → go to the right branch.
If No → go to the left branch.
Then another question is asked.
Eventually, the observation reaches a leaf, where the final prediction is made.

Main Components of a Decision Tree

1. Root Node
The first/top node of the tree.
It contains the first decision.
Example:
Income > 50?

2. Decision/Internal Node
A node where another decision is made.
Example:
Website Visits > 5?

3. Branch
A branch represents the outcome of a decision.
For example:

Income > 50?
/ \
Yes No

4. Leaf Node
The final node that gives the prediction.
Example:
Purchase
No Purchase
So:
Root → Decision Nodes → Branches → Leaf

Decision Tree for Classification

A Decision Tree Classifier is used when the target variable is categorical.

For example:
Purchase / No Purchase
Fraud / Genuine
Yes / No
Pass / Fail

Decision Tree for Regression

A Decision Tree can also predict continuous numerical values.
This is called a:
Regression Tree

For example:
Sales
Revenue
House price
Demand
Profit
In a regression tree, the leaf generally gives a numerical prediction based on the observations reaching that leaf.

What is CART?
CART stands for:
Classification and Regression Trees
It is a decision-tree methodology that can be used for both:
Classification
Regression

In Scikit-learn, we can implement these using:
DecisionTreeClassifier
and
DecisionTreeRegressor

How Does a Decision Tree Choose a Split?
Suppose the model has several possible variables:
Age
Income
Website visits
Previous purchases

The tree has to decide:
Which variable should be used for the next split?

It evaluates possible splits using an appropriate criterion.
For classification, one commonly used criterion is Gini impurity.

Gini Impurity
Gini impurity measures how mixed the classes are within a node.

Gini = 0 → node contains only one class
Higher Gini → classes are more mixed

What Happens During Tree Building?
The basic process is:

Start with all observations
↓
Find the best split
↓
Divide the data
↓
Find another best split
↓
Continue splitting
↓
Create leaf nodes
↓
Make predictions

What is Overfitting in a Decision Tree?

A decision tree can become very complicated if it keeps splitting until it almost perfectly fits the training data.

For example:
Training Accuracy = 100%
Testing Accuracy = 65%
This can indicate overfitting.

A very deep tree can memorize details of the training data rather than learning patterns that generalize well.

How Can We Control Overfitting?

One common method is:
max_depth

For example:
DecisionTreeClassifier(max_depth=3)
This limits the maximum depth of the tree.

Other parameters include:
min_samples_split
min_samples_leaf
max_leaf_nodes
These parameters help control tree complexity.

Advantages of Decision Trees
1. Easy to Understand
The decision rules can be explained to business users.
2. Easy to Visualize
The tree can be represented graphically.
3. Works with Different Variables
Decision trees can handle multiple predictor variables.
4. Useful for Business Decisions
The resulting rules can be directly useful for decision-making.
5. Can Perform Classification and Regression
Through CART, trees can handle both types of prediction problems.

Limitations
1. Overfitting
Deep trees can fit training data too closely.
2. Sensitive to Data Changes
Small changes in data can sometimes produce a different tree.
3. Large Trees Become Difficult to Interpret

A very deep tree can become complicated.

4. Greedy Splitting

The tree generally chooses the best split at each stage rather than searching all possible complete tree structures.

Business Applications

Decision trees are useful in:

Marketing

Predict whether a customer will purchase a product.

Banking

Assess loan approval.

Insurance

Predict claim categories.

Retail

Predict customer behaviour.

Sales

Predict sales or revenue using regression trees.